
Chapter 2: Algebra and Functions

A Level Pure Maths Notes · Rexponential

Why this chapter matters

Algebra and Functions is the toolkit chapter: almost nothing later in the course works without it. Coordinate geometry, calculus, trigonometric equations and sequences all lean on the skills here — factorising, manipulating inequalities, understanding what a function actually does to an input. If you are shaky on this chapter, you won't just struggle here; you'll struggle everywhere, because every later topic quietly assumes fluency with it.

Much of this chapter builds on GCSE, so the main challenge isn't meeting brand-new ideas — it's the jump in fluency and abstraction expected: from solving one quadratic to manipulating a family of quadratics in terms of an unknown constant, from “a function” as a vague rule to functions as objects you can combine, invert and compare.

How to use these notes

As in Chapter 1, each section has:

- **Concept** — what the idea means and why it works.
- **Procedure** — the exact method, as a numbered checklist.
- **Worked example(s)** — full questions with step-by-step solutions and indicative mark allocations.

A few topics in this chapter (partial fractions, the modulus function, the formal treatment of composite and inverse functions) are genuinely new material for most students rather than an extension of GCSE — these are flagged where they appear, since they're worth spending longer on.

2.1 Indices and surds

Concept

Indices (powers) are a shorthand for repeated multiplication, and the index laws are simply consequences of that definition — not rules to memorise in isolation. If you understand that $a^3 \times a^2$ means $(a \times a \times a) \times (a \times a) = a^5$, the law $a^m \times a^n = a^{m+n}$ stops being an arbitrary fact and becomes something you could rebuild from scratch if you forgot it.

A surd is an irrational root, like $\sqrt{2}$ or $\sqrt{5}$, left in exact form rather than rounded as a decimal. Working with surds matters because exact answers are frequently required at A level (a decimal answer where an exact one was asked for typically loses marks), and because surd manipulation underpins later work on coordinate geometry and trigonometry, where exact values appear constantly.

KEY FORMULAS: INDEX LAWS

- $a^m \times a^n = a^{m+n}$ (multiplying: add the powers)
- $a^m \div a^n = a^{m-n}$ (dividing: subtract the powers)
- $(a^m)^n = a^{mn}$ (a power of a power: multiply the powers)
- $a^0 = 1$ (anything to the power 0 is 1, since $a^n/a^n = a^0 = 1$)
- $a^{-n} = 1/a^n$ (a negative power means “reciprocal of”)
- $a^{(1/n)} = \sqrt[n]{a}$ (a fractional power with denominator n means “the nth root”)
- $a^{(m/n)} = (\sqrt[n]{a})^m = \sqrt[n]{(a^m)}$ (a general fractional power: take the root, then the power, in either order)

Procedure: simplifying surds

1. Look for a square factor inside the root (the largest one you can spot).
2. Split the root using $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$.
3. Take the square root of the square factor out of the root.

Procedure: rationalising the denominator

1. If the denominator is a single surd (like $\sqrt{5}$), multiply top and bottom by that surd.
2. If the denominator is of the form $a + \sqrt{b}$ or $a - \sqrt{b}$, multiply top and bottom by its conjugate ($a - \sqrt{b}$ or $a + \sqrt{b}$), which uses $(x + y)(x - y) = x^2 - y^2$ to remove the surd from the denominator.

Worked example 2.1a (index laws)
EXAMPLE 2.1A (4 MARKS)

Simplify $(8x^6)^{(2/3)}$, giving your answer in the form ax^b .

Step 1 Apply the power to each factor inside the bracket separately: $(8x^6)^{(2/3)} = 8^{(2/3)} \times (x^6)^{(2/3)}$. (M1)

Step 2 Evaluate $8^{(2/3)}$: this means $(\sqrt[3]{8})^2 = 2^2 = 4$. (M1 A1)

Step 3 Evaluate $(x^6)^{(2/3)}$ using the power-of-a-power rule: multiply $6 \times 2/3 = 4$, giving x^4 . (A1)

Answer Combine: the answer is $4x^4$. (final answer)

Worked example 2.1b (surds)
EXAMPLE 2.1B (5 MARKS)

(a) Simplify $\sqrt{72}$. (b) Write $5/\sqrt{3}$ in the form $k\sqrt{3}$, where k is a rational number.

Step 1 (a) Find the largest square factor of 72: $72 = 36 \times 2$. (M1)

Step 2 $\sqrt{72} = \sqrt{36} \times \sqrt{2} = 6\sqrt{2}$. (A1)

Step 3 (b) Multiply top and bottom by $\sqrt{3}$: $5/\sqrt{3} = (5\sqrt{3})/(\sqrt{3} \times \sqrt{3}) = (5\sqrt{3})/3$. (M1 A1)

Answer So $k = 5/3$, and the answer is $(5/3)\sqrt{3}$. (A1)

Worked example 2.1c (conjugate rationalising)

EXAMPLE 2.1C (4 MARKS)

Express $(4 + \sqrt{5})/(3 - \sqrt{5})$ in the form $(a + b\sqrt{5})/c$, where a , b and c are integers.

Step 1 Multiply top and bottom by the conjugate of the denominator, $3 + \sqrt{5}$. (M1)

Step 2 Numerator: $(4 + \sqrt{5})(3 + \sqrt{5}) = 12 + 4\sqrt{5} + 3\sqrt{5} + 5 = 17 + 7\sqrt{5}$. (M1)

Step 3 Denominator: $(3 - \sqrt{5})(3 + \sqrt{5}) = 9 - 5 = 4$. (A1)

Answer The answer is $(17 + 7\sqrt{5})/4$. (A1)

EXAM TIP

The single most common error in this section is applying an index law to a *sum* rather than a product — for example wrongly “simplifying” $(x + y)^2$ using an index law instead of expanding it. Remember the laws only work when terms are multiplied, divided, or raised to a power directly — never across a $+$ or $-$.

2.2 Quadratics

Concept

A quadratic expression $ax^2 + bx + c$ describes a parabola, and the three standard techniques — factorising, completing the square, and the quadratic formula — are three different routes to the same information: where the curve crosses the x -axis, and where its turning point is. Choosing the right tool matters: factorising is fastest when it works cleanly, completing the square directly reveals the turning point (and is essential for later topics like integration and circle equations), and the formula always works but gives the least structural insight.

The discriminant, $b^2 - 4ac$, is not a separate topic to memorise — it is simply the part of the quadratic formula under the square root. Its sign tells you how many real solutions exist because a negative number has no real square root: two solutions when it's positive, one repeated solution when it's zero, no real solutions when it's negative. Framing it this way (rather than as an isolated rule) helps you see why it works, not just what to do with it.

Procedure: completing the square

For $ax^2 + bx + c$:

1. Factor a out of the first two terms: $a(x^2 + (b/a)x) + c$.
2. Halve the coefficient of x inside the bracket, and write the bracket as a perfect square minus the correction term: $a[(x + b/2a)^2 - (b/2a)^2] + c$.
3. Expand the outer bracket and simplify the constant terms.
4. Read off the turning point directly: for $a(x + p)^2 + q$, the turning point is $(-p, q)$.

Procedure: the discriminant

1. Identify a , b and c from $ax^2 + bx + c = 0$.
2. Calculate $b^2 - 4ac$.

3. Interpret: > 0 means two distinct real roots; $= 0$ means one repeated real root (the curve touches the x-axis); < 0 means no real roots (the curve doesn't meet the x-axis).

Worked example 2.2a (completing the square)

EXAMPLE 2.2A (5 MARKS)

Express $2x^2 - 12x + 7$ in the form $a(x + p)^2 + q$, and state the coordinates of the minimum point.

Step 1 Factor 2 out of the first two terms: $2(x^2 - 6x) + 7$. (M1)

Step 2 Halve -6 to get -3 , and write $x^2 - 6x = (x - 3)^2 - 9$: $2[(x - 3)^2 - 9] + 7$. (M1 A1)

Step 3 Expand: $2(x - 3)^2 - 18 + 7 = 2(x - 3)^2 - 11$. (A1)

Step 4 Since $a = 2 > 0$, this is a minimum, at the point where $(x - 3)^2 = 0$, i.e. $(3, -11)$. (A1)

Answer $2(x - 3)^2 - 11$, minimum point $(3, -11)$.

Worked example 2.2b (discriminant, finding a range)

EXAMPLE 2.2B (5 MARKS)

Find the values of k for which the equation $x^2 + kx + (k + 8) = 0$ has two distinct real roots.

Step 1 For two distinct real roots, we need $b^2 - 4ac > 0$. (B1)

Step 2 Here $a = 1$, $b = k$, $c = k + 8$, so the condition is $k^2 - 4(k + 8) > 0$. (M1)

Step 3 Expand: $k^2 - 4k - 32 > 0$. (A1)

Step 4 Factorise: $(k - 8)(k + 4) > 0$. (M1)

Step 5 This is a positive quadratic in k , so it's greater than zero outside its roots: $k < -4$ or $k > 8$. (A1)

Answer $k < -4$ or $k > 8$.

EXAM TIP

In worked example 2.2b, the step from $(k - 8)(k + 4) > 0$ to the final range is exactly the quadratic-inequality skill from section 2.4 — this is a good moment to point out how sections in this chapter interlock rather than sit in isolation.

2.3 Simultaneous equations

Concept

Solving simultaneous equations means finding the point(s) where two graphs intersect. When both equations are linear, there is at most one intersection point, found by elimination or substitution. When one equation is linear and the other quadratic (or another curve), there can be zero, one or two intersection points — geometrically, a line can miss a curve, touch it once, or cross it twice. Keeping this graphical picture in mind helps you sanity-check your answers: two solutions should feel expected when a line crosses a parabola, and a single repeated solution corresponds to the line being a tangent.

Procedure: linear and non-linear (substitution method)

1. Rearrange the linear equation to make one variable the subject.
2. Substitute this expression into the non-linear equation, eliminating that variable.
3. Solve the resulting equation (usually a quadratic) for the remaining variable.
4. Substitute each solution back into the linear equation to find the paired value of the other variable.
5. Pair up the answers correctly — each x-value goes with its own y-value, not any combination.

Worked example 2.3a**EXAMPLE 2.3A (6 MARKS)**

Solve the simultaneous equations $y = 2x - 1$ and $x^2 + y^2 = 29$.

Step 1 The linear equation is already rearranged for y , so substitute directly: $x^2 + (2x - 1)^2 = 29$. (M1)

Step 2 Expand: $x^2 + 4x^2 - 4x + 1 = 29$, so $5x^2 - 4x - 28 = 0$. (M1 A1)

Step 3 Factorise: $(5x - 14)(x + 2) = 0$. (M1)

Step 4 So $x = 14/5$ or $x = -2$. (A1)

Step 5 Find the paired y -values: when $x = 14/5$, $y = 2(14/5) - 1 = 23/5$. When $x = -2$, $y = 2(-2) - 1 = -5$. (A1)

Answer $(14/5, 23/5)$ and $(-2, -5)$.

Worked example 2.3b (rearranging first)**EXAMPLE 2.3B (6 MARKS)**

Solve $x + 2y = 5$ and $x^2 - y^2 = 5$.

Step 1 Rearrange the linear equation: $x = 5 - 2y$. (M1)

Step 2 Substitute into the quadratic equation: $(5 - 2y)^2 - y^2 = 5$. (M1)

Step 3 Expand: $25 - 20y + 4y^2 - y^2 = 5$, so $3y^2 - 20y + 20 = 0$. (A1)

Step 4 Use the quadratic formula: $y = [20 \pm \sqrt{(400 - 240)}]/6 = [20 \pm \sqrt{160}]/6$. (M1)

Step 5 $\sqrt{160} = 4\sqrt{10}$, so $y = (20 \pm 4\sqrt{10})/6 = (10 \pm 2\sqrt{10})/3$. (A1)

Step 6 Substitute back to find each paired x -value using $x = 5 - 2y$. (A1)

Answer $y = (10 \pm 2\sqrt{10})/3$, with $x = 5 - 2y$ for each.

EXAM TIP

The step most people skip is substituting each solution back in to find its pair — leaving an answer as “ $x = 2$ or $-14/5$, $y = 3$ or $-33/5$ ” without the pairing shown is ambiguous and can lose marks, since it implies four possible combinations rather than two. Always write paired answers together, or as coordinate pairs.

2.4 Inequalities

Concept

An inequality describes a *range* of values rather than a single answer, and the algebra of inequalities is almost identical to that of equations — with one crucial exception: multiplying or dividing both sides by a negative number reverses the inequality sign. Students who treat inequalities exactly like equations, without watching for this, produce answers with the wrong direction.

For quadratic inequalities, the sketch of the parabola is not optional working — it's the fastest and most reliable way to read off the correct range, because it shows directly whether the required region is between the roots or outside them.

Procedure: linear inequalities

1. Treat it like an equation and rearrange to isolate the variable.
2. If you multiply or divide by a negative number at any stage, reverse the inequality sign.
3. Write the final answer in the form $x < / \leq / > / \geq$ a number (or as a combined range).

Procedure: quadratic inequalities

1. Rearrange so one side is zero.
2. Factorise (or use the quadratic formula) to find the critical values — the roots.
3. Sketch the parabola: it opens upward if the coefficient of x^2 is positive, downward if negative.
4. Read the required region off the sketch: “greater than zero” means where the curve is above the x-axis; “less than zero” means below it.
5. Write the answer using the critical values, taking care with strict ($<$, $>$) versus non-strict ($\leq$, $\geq$) inequalities.

Worked example 2.4a (linear)

EXAMPLE 2.4A (3 MARKS)

Solve $5 - 3x \geq 2x - 10$, and illustrate the solution on a number line.

Step 1 Collect x terms on one side: $5 + 10 \geq 2x + 3x$. (M1)

Step 2 Simplify: $15 \geq 5x$. (A1)

Step 3 Divide by 5 (positive, so no sign flip): $3 \geq x$, i.e. $x \leq 3$. (A1)

Answer $x \leq 3$.

Worked example 2.4b (quadratic)

EXAMPLE 2.4B (5 MARKS)

Solve $x^2 - 2x - 15 > 0$.

Step 1 Factorise: $(x - 5)(x + 3) > 0$. (M1 A1)

Step 2 The critical values are $x = 5$ and $x = -3$. (A1)

Step 3 Since the coefficient of x^2 is positive, the parabola opens upward and is positive (above the x-axis) outside the roots. (M1)

Step 4 So the solution is $x < -3$ or $x > 5$. (A1)

Answer $x < -3$ or $x > 5$.

Worked example 2.4c (a trickier quadratic inequality)

EXAMPLE 2.4C (5 MARKS)

Solve $12 - x - x^2 \geq 0$.

Step 1 It's easier to work with a positive leading coefficient, so multiply through by -1 , remembering to reverse the inequality: $x^2 + x - 12 \leq 0$. (M1)

Step 2 Factorise: $(x + 4)(x - 3) \leq 0$. (M1 A1)

Step 3 Critical values $x = -4$ and $x = 3$; the parabola opens upward, so it's ≤ 0 between the roots. (M1)

Step 4 Solution: $-4 \leq x \leq 3$. (A1)

Answer $-4 \leq x \leq 3$.

EXAM TIP

Worked example 2.4c is exactly where the sign-reversal rule matters most — students who forget to flip the inequality when multiplying by -1 will factorise correctly but state the region on the wrong side of the roots. Always sketch the graph as a check: does the shape of the parabola match the direction of the final answer?

2.5 Polynomials: the factor and remainder theorems

Concept

The remainder theorem says that when a polynomial $f(x)$ is divided by $(x - a)$, the remainder equals $f(a)$. This is worth deriving rather than just memorising: if $f(x) = (x - a) \times Q(x) + R$ for some quotient $Q(x)$ and remainder R , then substituting $x = a$ makes the $(x - a)$ term vanish, leaving $f(a) = R$. Once you see why it's true, it stops being a magic shortcut and becomes an obvious consequence of what division means.

The factor theorem is the special case where the remainder is zero: if $f(a) = 0$, then $(x - a)$ is a factor of $f(x)$. This turns “find the factors of a cubic” from guesswork into a search strategy: test values of a (usually factors of the constant term) until $f(a) = 0$ is found, then divide to find the rest.

Procedure

1. To test whether $(x - a)$ is a factor, substitute $x = a$ into $f(x)$. If $f(a) = 0$, it's a factor (factor theorem).
2. To find a first factor of a cubic (or higher) with no given root, try small integer values that divide the constant term (± 1 , ± 2 , ± 3 , and so on).
3. Once a linear factor is found, divide the polynomial by it (using algebraic long division, or by comparing coefficients) to find the remaining quadratic factor.
4. Factorise the remaining quadratic normally, or use the quadratic formula if it doesn't factorise.

Worked example 2.5a (remainder theorem)

EXAMPLE 2.5A (3 MARKS)

Find the remainder when $f(x) = x^3 - 4x^2 + 2x + 5$ is divided by $(x - 2)$.

Step 1 By the remainder theorem, the remainder is $f(2)$. (M1)

Step 2 $f(2) = 8 - 16 + 4 + 5 = 1$. (A1 — ft, arithmetic follow-through)

Answer The remainder is 1. (A1)

Worked example 2.5b (factor theorem, finding an unknown constant)

EXAMPLE 2.5B (4 MARKS)

Given that $(x + 1)$ is a factor of $f(x) = 2x^3 + 3x^2 + kx - 4$, find the value of k .

Step 1 Since $(x + 1)$ is a factor, $f(-1) = 0$ by the factor theorem. (M1)

Step 2 $f(-1) = 2(-1)^3 + 3(-1)^2 + k(-1) - 4 = -2 + 3 - k - 4 = -3 - k$. (A1 — ft, correct simplification)

Step 3 Set this equal to zero: $-3 - k = 0$. (A1)

Answer So $k = -3$. (A1)

Worked example 2.5c (full factorisation of a cubic)**EXAMPLE 2.5C (7 MARKS)**

$f(x) = x^3 - 2x^2 - 5x + 6$. (a) Show that $(x - 1)$ is a factor. (b) Hence factorise $f(x)$ completely.

Step 1 (a) $f(1) = 1 - 2 - 5 + 6 = 0$. Since $f(1) = 0$, $(x - 1)$ is a factor by the factor theorem. (M1 A1)

Step 2 (b) Divide $f(x)$ by $(x - 1)$. The quotient must be a quadratic, and its constant term can be found without doing full long division: since $(x - 1)(x^2 + ax + c)$ must have constant term 6, and $(-1) \times c = 6$ gives $c = -6$, we can write $x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 + ax - 6)$ for some a to find. (M1)

Step 3 Compare the x^2 coefficient on both sides: expanding $(x - 1)(x^2 + ax - 6)$ gives an x^2 term of $(a - 1)x^2$, so $a - 1 = -2$, giving $a = -1$. This gives $x^3 - 2x^2 - 5x + 6 = (x - 1)(x^2 - x - 6)$. (M1 A1)

Step 4 Factorise the quadratic: $x^2 - x - 6 = (x - 3)(x + 2)$. (M1)

Answer So $f(x) = (x - 1)(x - 3)(x + 2)$. (A1)

EXAM TIP

For step 3 (comparing coefficients), you can also compare the constant term as a check: $(-1)(-6) = 6$, which matches $f(x)$'s constant term — a quick way to catch an arithmetic slip before moving on.

2.6 Partial fractions

This is new material for most students — it doesn't build directly on a GCSE skill, so allow yourself extra time.

Concept

Partial fractions reverses the process of combining fractions over a common denominator. A single complicated fraction like $(7x + 4)/((x + 1)(x - 2))$ can be split back into two simpler fractions, $A/(x + 1) + B/(x - 2)$, that add up to the original. This matters because many later techniques — particularly integration — can only be applied to simple fractions, not to a combined one; partial fractions is the tool that makes a complicated fraction tractable.

The form the split fractions take depends on the type of factor in the denominator: a simple linear factor gets a constant numerator ($A/(x + 1)$); a repeated linear factor needs two terms, one for each power ($A/(x + 1) + B/(x + 1)^2$); and if the original fraction is improper (the top has an equal or higher degree than the bottom), a polynomial term must be added before splitting.

Procedure

1. Check the degree of the numerator against the denominator. If the fraction is improper, divide first to get a polynomial plus a proper fraction, then apply the rest of the method to the proper fraction.
2. Write the denominator's factors and set up the partial fraction form (constants for simple factors; two terms for a repeated factor).

3. Multiply both sides by the original denominator to clear all fractions, giving a polynomial identity.
4. Find the unknown constants either by substituting convenient values of x (values that make one bracket zero) or by comparing coefficients.

Worked example 2.6a (distinct linear factors)

EXAMPLE 2.6A (5 MARKS)

Express $(9x - 2)/((x - 1)(2x + 1))$ in partial fractions.

Step 1 Set up the form: $(9x - 2)/((x - 1)(2x + 1)) = A/(x - 1) + B/(2x + 1)$. (M1)

Step 2 Multiply both sides by $(x - 1)(2x + 1)$: $9x - 2 = A(2x + 1) + B(x - 1)$. (M1)

Step 3 Substitute $x = 1$ (this makes the B term vanish): $9(1) - 2 = A(3)$, so $A = 7/3$. (M1 A1)

Step 4 Substitute $x = -1/2$ (this makes the A term vanish): $9(-1/2) - 2 = B(-3/2)$, so $-13/2 = -3B/2$, giving $B = 13/3$. (A1)

Answer $A = 7/3$, $B = 13/3$.

Worked example 2.6b (repeated linear factor)

EXAMPLE 2.6B (6 MARKS)

Express $(5x + 4)/(x(x + 2)^2)$ in partial fractions.

Step 1 Set up the form with two terms for the repeated factor: $(5x + 4)/(x(x + 2)^2) = A/x + B/(x + 2) + C/(x + 2)^2$. (M1)

Step 2 Multiply through by $x(x + 2)^2$: $5x + 4 = A(x + 2)^2 + Bx(x + 2) + Cx$. (M1)

Step 3 Substitute $x = 0$: $4 = A(4)$, so $A = 1$. (M1 A1)

Step 4 Substitute $x = -2$ (this makes the A and B terms vanish): $-10 + 4 = C(-2)$, so $-6 = -2C$, giving $C = 3$. (A1 — ft, same method as before)

Step 5 Compare the coefficient of x^2 (which must be zero, since there's no x^2 term on the left): $0 = A + B$, so $B = -1$. (A1)

Answer $A = 1$, $B = -1$, $C = 3$.

EXAM TIP

In worked example 2.6b, once the “easy” substitutions ($x = 0$ and $x = -2$) are used up, the remaining constant needs a different technique — comparing coefficients of the highest power is usually the fastest. Always find as many constants as possible by substitution first, since it's less error-prone, and only fall back to comparing coefficients for what's left.

2.7 Graph transformations

Concept

A transformation takes the graph of $y = f(x)$ and moves, stretches or reflects it, without needing to know what f actually is. This is a powerful idea precisely because it's general — the same four rules work whether $f(x)$ is a quadratic, a trig function, or something entirely unfamiliar.

The genuinely tricky part is that changes *inside* the function (affecting x) behave the opposite way to how they look, while changes *outside* the function (affecting y) behave exactly as they look. This is because a transformation affecting x asks “what input now produces the same output as before” — which reverses the intuitive direction.

For a graph $y = f(x)$:

KEY FORMULAS: THE FOUR STANDARD TRANSFORMATIONS

- $y = f(x) + a$ — translation by a in the y -direction (moves up if $a > 0$). This is “outside”, so it behaves as expected.
- $y = f(x + a)$ — translation by $-a$ in the x -direction (moves *left* if $a > 0$). This is “inside”, so the direction is reversed from what looks intuitive.
- $y = a f(x)$ — vertical stretch, scale factor a , from the x -axis. “Outside”, so it behaves as expected (values get bigger).
- $y = f(ax)$ — horizontal stretch, scale factor $1/a$, from the y -axis. “Inside”, so the graph gets squashed when $a > 1$, not stretched.

Procedure: applying a transformation to a known point

1. Identify the coordinates of the known point on $y = f(x)$.
2. Work out how x must change to give the same output under the new function (for “inside” changes), or how y changes directly (for “outside” changes).
3. Write the new coordinates.

Worked example 2.7a**EXAMPLE 2.7A (4 MARKS)**

The curve $y = f(x)$ has a maximum point at $(3, 5)$. State the coordinates of the maximum point on each of the following: (a) $y = f(x) - 2$, (b) $y = f(x - 4)$, (c) $y = 2f(x)$, (d) $y = f(2x)$.

Step 1 (a) Outside change, shifts y down by 2: $(3, 3)$. (B1)

Step 2 (b) Inside change: $f(x - 4)$ means the graph moves right by 4 (opposite of the sign shown): $(7, 5)$. (B1)

Step 3 (c) Outside change, y -values doubled: $(3, 10)$. (B1)

Step 4 (d) Inside change: to get the same output as f at $x = 3$, we now need $2x = 3$, so $x = 1.5$. The y -value is unchanged: $(1.5, 5)$. (B1)

Answer (a) $(3, 3)$ (b) $(7, 5)$ (c) $(3, 10)$ (d) $(1.5, 5)$.

Worked example 2.7b (combined and applied to an equation)**EXAMPLE 2.7B (5 MARKS)**

The graph of $y = f(x)$ passes through the point $(-2, 0)$ and has a root at $x = -2$. Given $g(x) = f(x + 3) - 1$, find the coordinates of the corresponding point on $y = g(x)$.

Step 1 Identify the two transformations: $f(x + 3)$ is an inside change (shift left by 3), and -1 is an outside change (shift down by 1). (B1)

Step 2 Apply the horizontal shift first to the x-coordinate: the point on f at $x = -2$ corresponds to g at $x = -2 - 3 = -5$ (subtract 3, since shifting the graph left by 3 means each feature now occurs 3 earlier). (M1 A1)

Step 3 Apply the vertical shift to the y-coordinate: $y = 0 - 1 = -1$. (M1)

Answer The corresponding point on $g(x)$ is $(-5, -1)$. (A1)

EXAM TIP

For worked example 2.7b step 2, a very common error is adding 3 instead of subtracting — substituting a test value into $g(x + 3) = f((-5) + 3) = f(-2)$ as a direct check is a reliable way to catch this before it becomes a habit.

2.8 Functions: domain, range, composite and inverse functions

The formal treatment here is largely new — you may have used function notation loosely before, but not with this precision.

Concept

A function is a rule that assigns exactly one output to each valid input — the “exactly one” is essential: if a rule could give two different outputs for the same input, it isn't a function. The **domain** is the set of inputs the function is allowed to take, and the **range** is the set of outputs it actually produces. These are not the same thing, and mixing them up is one of the most common errors at this level: the domain is about what you're allowed to put in; the range is about what comes out.

A **composite function**, written $fg(x)$, means “do g first, then do f to the result” — reading order matters and is the opposite of how it's written; $fg(x)$ is not the same as $gf(x)$ in general. An **inverse function**, $f^{-1}(x)$, undoes what f does: if f takes 3 to 7, then f^{-1} takes 7 back to 3. Not every function has an inverse that is itself a function — only a one-to-one function does, because otherwise the inverse would need to send one input to multiple outputs, which breaks the definition of a function.

Procedure: finding the range

1. Consider what the domain restricts the inputs to.
2. Think about the shape of the graph (or complete the square for a quadratic) to see the greatest and least output values reached.
3. Write the range using inequality notation, matching the style of the domain.

Procedure: composite functions

1. To find $fg(x)$, first work out $g(x)$, then substitute that whole expression into f in place of x .

2. Always work from the inside outward: the function closest to x is applied first.

Procedure: inverse functions

1. Write $y = f(x)$.
2. Rearrange the equation to make x the subject.
3. Swap x and y (or simply relabel): the resulting expression for y (in terms of x) is $f^{-1}(x)$.
4. The domain of f^{-1} is the range of f , and the range of f^{-1} is the domain of f — they swap.

Worked example 2.8a (domain and range)

EXAMPLE 2.8A (4 MARKS)

$f(x) = (x - 2)^2 + 3$ for $x \geq 2$. State the range of f .

Step 1 The function is already in completed-square form: the minimum value occurs at $x = 2$, where $f(2) = 3$. (M1)

Step 2 Since the domain is restricted to $x \geq 2$ (only the right-hand branch of the parabola), the function is increasing throughout the domain. (M1)

Step 3 So the smallest output is 3, and outputs increase without bound as x increases. (A1)

Answer Range: $f(x) \geq 3$. (A1)

Worked example 2.8b (composite and inverse)

EXAMPLE 2.8B (7 MARKS)

$f(x) = 3x + 2$ and $g(x) = x^2 - 1$ for $x \in \mathbb{R}$. (a) Find $fg(2)$. (b) Find $gf(x)$ in terms of x . (c) Find $f^{-1}(x)$.

Step 1 (a) Work inside-out: $g(2) = 4 - 1 = 3$. Then $f(3) = 3(3) + 2 = 11$. (M1 A1)

Step 2 (b) $gf(x)$ means apply f first: $f(x) = 3x + 2$. Then apply g to this: $g(3x + 2) = (3x + 2)^2 - 1$. (M1)

Step 3 Expand: $(3x + 2)^2 - 1 = 9x^2 + 12x + 4 - 1 = 9x^2 + 12x + 3$. (A1)

Step 4 (c) Let $y = 3x + 2$. Rearrange to make x the subject: $y - 2 = 3x$, so $x = (y - 2)/3$. (M1 A1)

Answer (a) 11 (b) $9x^2 + 12x + 3$ (c) $f^{-1}(x) = (x - 2)/3$. (A1)

EXAM TIP

Students often try to find $f^{-1}(x)$ by swapping x and y in the ORIGINAL equation before rearranging, which usually leads to more complicated algebra than necessary. Use rearrange-then-relabel as the standard order, and always do a quick check: does $ff^{-1}(x) = x$?

2.9 The modulus function

New material — there's no GCSE equivalent, so treat this as a fresh topic rather than an extension.

Concept

The modulus of a number, $|x|$, is its distance from zero, always non-negative: $|5| = 5$ and $|-5| = 5$. Graphically, $y = |f(x)|$ is formed by taking the graph of $y = f(x)$ and reflecting any part that dips below

the x-axis up above it — the whole graph ends up on or above the x-axis. This picture is the single most useful thing to have before any algebra: once you can sketch $|f(x)|$ from $f(x)$, modulus equations and inequalities become much more intuitive to check.

When solving an equation involving a modulus, the key idea is that $|A| = B$ (for $B \geq 0$) means $A = B$ **or** $A = -B$ — the expression inside the modulus could have been positive or negative before the modulus was applied, so both cases must be considered and checked.

Procedure: solving modulus equations

1. If the modulus is isolated on one side ($|A| = B$), split into two cases: $A = B$ and $A = -B$.
2. Solve each resulting equation normally.
3. Check each solution in the *original* equation — not every algebraic solution is valid, since squaring or splitting can introduce extra answers that don't actually satisfy the original modulus equation.

KEY FORMULAS: SOLVING MODULUS INEQUALITIES

- For $|A| < k$ ($k > 0$): this means A is within k of zero, so $-k < A < k$.
- For $|A| > k$ ($k > 0$): this means A is further than k from zero, so $A < -k$ or $A > k$.

Worked example 2.9a (equation, both sides could be modulus)

EXAMPLE 2.9A (5 MARKS)

Solve $|2x - 3| = |x + 4|$.

Step 1 Split into two cases, since either expression could be negated: Case 1: $2x - 3 = x + 4$.

Case 2: $2x - 3 = -(x + 4)$. (M1)

Step 2 Case 1: $2x - 3 = x + 4$ gives $x = 7$. (M1 A1)

Step 3 Case 2: $2x - 3 = -x - 4$ gives $3x = -1$, so $x = -1/3$. (M1 A1)

Answer Check both in the original equation: at $x = 7$, $|11| = |11|$ ✓. At $x = -1/3$, $|-11/3| = |11/3|$ ✓. Both solutions are valid.

Worked example 2.9b (inequality)

EXAMPLE 2.9B (4 MARKS)

Solve $|3x + 1| \leq 7$.

Step 1 This is of the form $|A| \leq k$, so $-7 \leq 3x + 1 \leq 7$. (M1)

Step 2 Subtract 1 from all three parts: $-8 \leq 3x \leq 6$. (M1)

Answer Divide by 3: $-8/3 \leq x \leq 2$. (A1 A1)

Worked example 2.9c (equation needing a check that rejects a solution)

EXAMPLE 2.9C (5 MARKS)

Solve $|2x - 1| = 3x - 4$.

Step 1 Split into two cases. Case 1: $2x - 1 = 3x - 4$, giving $x = 3$. (M1 A1)

Step 2 Case 2: $2x - 1 = -(3x - 4) = -3x + 4$, giving $5x = 5$, so $x = 1$. (M1 A1)

Step 3 Check both: at $x = 3$, LHS = $|5| = 5$, RHS = $9 - 4 = 5$ ✓. At $x = 1$, LHS = $|1| = 1$, RHS = $3 - 4 = -1$ ✗ (a modulus can never equal a negative number). (A1)

Answer So the only valid solution is $x = 3$; $x = 1$ is rejected.

EXAM TIP

Worked example 2.9c is exactly why the checking step is not optional — the algebra of Case 2 is entirely correct, but the resulting value doesn't satisfy the original equation because the right-hand side becomes negative while a modulus cannot. Make checking against the original equation a non-negotiable last step for every modulus equation, not just good practice.

Chapter summary

Section	Core skill	Watch out for
2.1 Indices and surds	Applying index laws; rationalising denominators	Index laws only apply across products, never across + or –
2.2 Quadratics	Completing the square; discriminant	Sign of the discriminant tells you the <i>number</i> of roots, not their value
2.3 Simultaneous equations	Substitution into a quadratic	Pairing each x with its correct y
2.4 Inequalities	Sketching to find the correct region	Flipping the inequality when multiplying/dividing by a negative
2.5 Polynomials	Factor and remainder theorems	Comparing coefficients correctly after dividing
2.6 Partial fractions	Splitting a fraction by its factor types	Repeated factors need two terms, not one
2.7 Graph transformations	Inside vs outside changes	Inside changes (affecting x) act in the opposite direction to how they look

Section	Core skill	Watch out for
2.8 Functions	Domain/range, composite, inverse	Domain $\neq$ range; $fg(x)$ means g first, then f
2.9 Modulus function	Splitting into two cases; checking solutions	A modulus can never equal a negative number — always check

Common exam mistakes across this chapter

- **Confusing domain and range.** A quick habit to build: domain is what goes in (read along the x -axis), range is what comes out (read along the y -axis).
- **Reversed transformation direction.** Any change made to x directly (inside the function) moves the graph the opposite way from what it looks like it should.
- **Not checking modulus solutions.** Every algebraic solution to a modulus equation must be verified against the original equation, since the case-splitting process can generate invalid extra solutions.
- **Forgetting to reverse an inequality sign** when multiplying or dividing by a negative number.
- **Treating a repeated factor in partial fractions like a distinct one**, missing the required extra term.

Suggested practice

Because this chapter's skills recur constantly later in the course, it's worth interleaving practice rather than moving straight on once each section is "done" — a short weekly set of 4–5 mixed questions drawn from earlier sections (alongside whatever is currently being taught) will do more for retention than a single block of practice at the end of the chapter.

You should now be able to

- Manipulate indices and surds, and solve quadratics by factorising, completing the square or the formula
- Solve linear–quadratic simultaneous equations and linear or quadratic inequalities
- Apply the factor and remainder theorems, and split an algebraic fraction into partial fractions
- Sketch transformed graphs, and work with domain, range, composite and inverse functions, and the modulus function